

Quick Guide: Industrial Laser Distance and Level Sensor (RS485/Modbus/0-20 & 4-20 mA)

1. Main Technical Parameters

- **Measurement range:** 0.2...8.0 m (hardware receiver blind zone: 200 mm).
- **Measurement accuracy:** ±6 cm in the range 0.2...3.0 m; ±2% of measured value in the range 3.0...8.0 m.
- **Measurement resolution:** 1 mm (interpolated).
- **Sampling frequency:** Configurable, 1...100 Hz (default: 10 Hz).
- **Supply voltage:** 8...28 VDC (12 VDC recommended).
- **Power consumption:** ≤ 3 W.
- **Communication interface:** RS-485 (half-duplex, Modbus RTU, default port settings: 9600 bps, 8N1).
- **Analog output:** Active 4–20 mA / 0–20 mA current loop (12-bit DAC, load resistance ≤ 250 Ω).
- **Analog output modes:** 0 – Distance 4–20 mA, 1 – Level 4–20 mA, 2 – Distance 0–20 mA, 3 – Level 0–20 mA, 4 – Discrete Alarm.
- **Non-volatile memory:** EEPROM (≥ 100,000 rewrite cycles).

2. Electrical Connection (Connector Pinout and Wire Colors)

Perform electrical wiring only when power is disconnected. Connect using an screened cable (shield grounded at the control cabinet side). For lines exceeding 150 m, connect a 120 Ω (0.25 W) termination resistor across lines A and B.

Pin	Wire Color	Signal Name	Purpose
1	Red	VCC	Power supply plus (+8...28 VDC, 12 VDC rec.)
2	Blue	GND	Power supply minus / Common (-)
3	Black	RS485 - A	RS-485 non-inverting line (A)
4	Yellow	RS485 - B	RS-485 inverting line (B)
5	Green	A0+	Active current loop output (+)
6	White	A0-	Active current loop output (-) / Common GND

3. Complete Modbus Register Map (Holding Registers)

Data exchange uses standard Modbus RTU functions: 03h (read holding registers), 06h (write single register), and 10h (write multiple registers). Changes take effect immediately upon writing.

Register	Parameter Name	Access	Type	Range / Default / Description
0	Distance	R	UInt16	0...12000 mm. Filtered target distance.

Register	Parameter Name	Access	Type	Range / Default / Description
1	Signal Strength	R	UInt16	0...65535. Reflected IR amplitude (<100 indicates weak signal).
2	Sensor Temperature	R	Int16	-40...125 °C. Internal sensor temperature.
3	Analog Output Current	R	UInt16	0...2000 (hundredths of mA, e.g., 1254 = 12.54 mA).
4	Fill Level	R	UInt16	0...100%. Calculated bin fill percentage.
5	Discrete Status	R	UInt16	0 = Normal, 1 = Overfill threshold exceeded.
6	System Error Status	R	UInt16	0 = OK; Bit 0 (0x01) = Rangefinder Watchdog timeout; Bit 1 (0x02) = DAC I ² C error; 3 = Multiple fault.
7...9	Reserved	R	UInt16	0. Block read alignment registers.
10	Modbus Address	R/W	UInt16	1...247 (Default: 1). Network address.
11	Sampling Frequency	R/W	UInt16	1...100 Hz (Default: 10). Measurement trigger rate.
12	Median Filter Window	R/W	UInt16	1...15 samples (Default: 10, 1 = Filter OFF).
13	EMA Smoothing	R/W	UInt16	0...100% (Default: 20%). Exponential moving average coefficient.
14	Calibration Point D1	R/W	UInt16	0...12000 mm (Default: 500). Reference distance #1.
15	Calibration Point D2	R/W	UInt16	0...12000 mm (Default: 5000). Reference distance #2.
16	Distance Boundary X1	R/W	UInt16	0...12000 mm (Default: 250). Distance corresponding to min current.
17	Distance Boundary X2	R/W	UInt16	0...12000 mm (Default: 5000). Distance corresponding to max current.
18	DAC Code Y1	R/W	UInt16	0...4095 (Default: 819). Raw DAC calibration code for min current (4 mA).
19	DAC Code Y2	R/W	UInt16	0...4095 (Default: 4095). Raw DAC calibration code for max current (20 mA).
20	Service/Mode Control	R/W	UInt16	0 = Normal; 1 = Test min current; 2 = Test max current; 3 = Capture M1; 4 = Capture M2 & Calculate; 5 = Capture D_empty; 6 = Capture D_full.
21	Empty Bin D_empty	R/W	UInt16	0...12000 mm (Default: 5000). Distance to empty bin bottom (0%).

Register	Parameter Name	Access	Type	Range / Default / Description
22	Full Bin D_full	R/W	UInt16	0...12000 mm (Default: 500). Distance to full bin top (100%).
23	Analog Output Mode	R/W	UInt16	0 = Distance 4–20 mA; 1 = Level 4–20 mA; 2 = Distance 0–20 mA; 3 = Level 0–20 mA; 4 = Discrete Alarm. (Default: 0).
24	Alarm Threshold	R/W	UInt16	10...100% (Default: 90). Fill percentage threshold for discrete alert.
25	Alarm Hysteresis	R/W	UInt16	1...50% (Default: 5). Reset zone for alarm status.
26	Alarm Current (Log 0)	R/W	UInt16	0...20 mA (Default: 0). Output current for Normal state in Mode 4.
27	Alarm Current (Log 1)	R/W	UInt16	0...20 mA (Default: 20). Output current for Alarm state in Mode 4.
28	Power-on Startup Current	R/W	UInt16	0...20 mA (Default: 0). Initial current hardware output upon power-up.

4. Calibration Procedures

4.1 Two-Point Sensor Calibration (D1/D2)

1. Place a flat target at a precisely known distance (e.g., 500 mm). Write this value to `Reg14 = 500`.
2. Send point 1 capture command: write `Reg20 = 3`.
3. Move target to the second known distance (e.g., 4000 mm). Write this value to `Reg15 = 4000`.
4. Send point 2 capture command: write `Reg20 = 4`. The MCU will compute scaling factors, store them in EEPROM, and automatically reset `Reg20` to `0`.

Calibration Guard Note:

If the two-point procedure is interrupted before issuing command `4`, all interim raw samples are discarded, and previous valid calibration factors remain active.

4.2 Container / Bunker Boundaries Calibration

- **Empty Bin (0%):** Completely clear the container. Direct the laser beam at the bottom. Write `Reg20 = 5`. The current distance is saved to `Reg21` (D_empty), and `Reg20` resets to `0`.
- **Full Bin (100%):** Fill the bin to maximum operating level. Write `Reg20 = 6`. The current distance is saved to `Reg22` (D_full), and `Reg20` resets to `0`.

Safety Protection: The MCU enforces a minimum 10 mm delta between boundaries (`D_empty > D_full + 10 mm`). If incorrect parameters are written or captured, dependent values automatically shift to prevent division-by-zero errors.

5. Configuration and Calibration of the Analog Output

5.1 Output Mode Selection (Reg23)

- **0** — **Distance 4–20 mA**: Scales linearly between **Reg16** (X1 = 4 mA) and **Reg17** (X2 = 20 mA).
- **1** — **Level 4–20 mA**: Proportional to fill level (0% at D_empty = 4 mA; 100% at D_full = 20 mA).
- **2** — **Distance 0–20 mA**: Scales linearly between **Reg16** (X1 = 0 mA) and **Reg17** (X2 = 20 mA).
- **3** — **Level 0–20 mA**: Proportional to fill level (0% = 0 mA; 100% = 20 mA).
- **4** — **Discrete Alarm**: Outputs **Reg26** current (default 0 mA) when **Reg5 = 0**, and **Reg27** current (default 20 mA) when **Reg5 = 1**.

5.2 Current Loop Hardware Trimming (Y1 / Y2)

1. Connect a milliammeter in series across **A0+** and **A0-** (loop load resistance $\leq 250 \Omega$).
2. **4.00 mA Point Trimming**: Write **Reg20 = 1**. Adjust raw code in **Reg18** (Y1) until the meter reads exactly **4.00 mA**. Write **Reg20 = 0** to save to EEPROM.
3. **20.00 mA Point Trimming**: Write **Reg20 = 2**. Adjust raw code in **Reg19** (Y2) until the meter reads exactly **20.00 mA**. Write **Reg20 = 0** to save to EEPROM.

5.3 Connecting Discrete Output to Controller Logic (Arduino/PLC)

When operating in Mode 4 (Discrete Alarm), convert current to voltage using a shunt resistor (R_{shunt}):

$$R_{shunt} (\Omega) = (U_{ctrl} \cdot 1000) / I \text{ (mA)}$$

- **5V TTL Logic (Arduino/Relay, 20 mA)**: Use a **250 Ω** resistor (0.25 W). **Reg26 = 0 mA**, **Reg27 = 20 mA**.
- **3.3V Logic (ESP32/STM32, 20 mA)**: Use a **165 Ω** resistor (0.25 W). **Reg26 = 0 mA**, **Reg27 = 20 mA**.

6. Physical ToF Principles & Installation Requirements

- **Blind Zone**: Objects closer than **200 mm** from the front glass saturate the receiver and return **0 mm**. Mount the device with at least 220–250 mm clearance above maximum fill height.
- **Target Reflectivity**: White/light surfaces (90% reflectivity) achieve maximum **8.0 m** range. Dark/black materials (10% reflectivity, e.g., coal, dark wet soil) reduce maximum range to **2.5 m**.
- **Beam Divergence (FoV)**: 2° total cone angle. Spot diameter increases by ~3.5 cm per meter of distance ($d \approx 0.035 \times D$). At 8 m, spot diameter is 28 cm. Ensure clearance from internal ladders or structural beams.
- **Ambient Light Immunity**: Protected up to **70 Klux**. Avoid direct sunlight falling straight into the receiver lens; install a visor for outdoor use.
- **Special Media**: For dusty environments, set EMA smoothing (**Reg13**) to 40...80%. For specular targets (mirrors, polished metal, clean quiet liquid), mount beam strictly at 90° or use a floating diffuser target.

7. Troubleshooting & Factory Reset

Symptom	Probable Cause	Remedy
No Modbus response	Power supply failure / A-B wires reversed / Modbus address mismatch	Verify 8...28 VDC at pins 1 & 2; swap wires on pins 3 & 4; scan address range or query default address 1 at 9600 8N1.
Distance frozen at 0 mm	Target inside blind zone (< 200 mm)	Move sensor further from surface to meet the minimum 200 mm operational clearance.
Distance reading fluctuates	Surface turbulence / Obstacles in 2° FoV cone	Increase median window (Reg12 to 12...15) and EMA smoothing (Reg13 to 30...50%). Re-aim sensor away from structure walls.
Fill level (Reg4) stays at 0%	D_empty and D_full not calibrated or invalid ($D_empty \leq D_full$)	Perform container boundary calibration (Section 4.2). Ensure D_empty is greater than D_full.
Current fixed at 4 or 20 mA	Device stuck in DAC test mode	Write 0 to control register Reg20 .
Reg 6 indicates Error Bit 1 (Value 2)	I ² C/DAC hardware failure or line fault	Check PCB traces, soldering of DAC IC, and absence of shorts on SDA/SCL lines. Recalibrate current output.

7.1 Factory Reset Values

To restore default parameters, write the following values to registers:

[Reg10](#)=1 , [Reg11](#)=10 , [Reg12](#)=10 , [Reg13](#)=20 , [Reg14](#)=500 , [Reg15](#)=5000 , [Reg16](#)=250 , [Reg17](#)=5000 , [Reg18](#)=819 , [Reg19](#)=4095 , [Reg21](#)=5000 , [Reg22](#)=500 , [Reg23](#)=0 , [Reg24](#)=90 , [Reg25](#)=5 , [Reg26](#)=0 , [Reg27](#)=20 , [Reg28](#)=0 .

Reboot power after applying factory reset values to re-initialize hardware filters.